

EC466/966: Estimation and Inference in Econometrics

Exercise # 3

1. Please answer the questions on EC966_466hw3.py. Please copy and paste **hprice.xls** file in the relevant folder on your computer.
2. Suppose that you have the following regression model

$$\mathbf{y} = \mathbf{X}\beta + \epsilon,$$

where $\mathbf{y}$ is the $n \times 1$ vector of the dependent variable, $\mathbf{X}$ is the $n \times K$ vector of regressors, β is the $K \times 1$ vector of the model parameters, and ϵ is the $n \times 1$ vector of the error term. Assume that Classical Linear Regression Model assumptions hold except the homoskedasticity assumption, i.e., $E(\epsilon\epsilon'|\mathbf{X}) = \mathbf{V}$ where $\mathbf{V}$ is a symmetric and positive definite matrix, and is a function of $\mathbf{X}$.

- (a) Derive the Generalized Least Squares (GLS) estimator.
- (b) Show that the GLS is an (conditionally) unbiased estimator.
- (c) Derive the variance-covariance matrix for the GLS estimator.
- (d) Is the OLS estimator unbiased in the presence of heteroskedasticity?